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Radical Acylfluoroalkylation of 1,3-Enynes via *N*-Heterocyclic Carbene/Photoredox Cooperative Catalysis

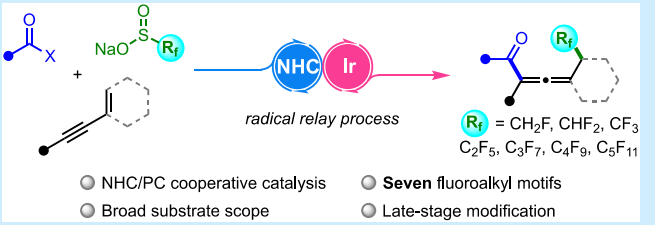
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ABSTRACT: We report a novel three-component radical acylfluoroalkylation of 1,3-enynes by synergistic *N*-heterocyclic carbene (NHC)/photoredox catalysis toward various fluorinated allenic aryl ketones. This protocol features a broad substrate scope and excellent functional group tolerability, with examples of late-stage modification of drug molecules and natural products. Notably, seven different fluoroalkyl motifs can be introduced to 1,3-enynes, further demonstrating the robustness and generality of this method. The generation of the fluoroalkyl radical from each sulfinate reagent was individually supported by EPR experiments.



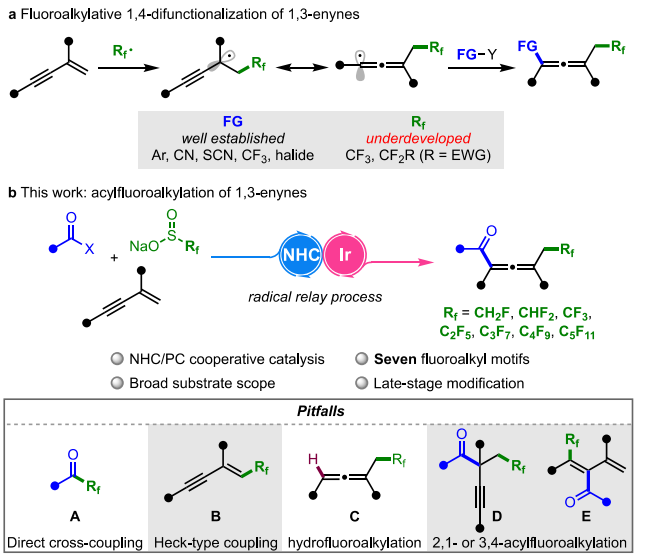
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Fluorine-containing molecules play a vital role in pharmaceuticals and agrochemicals.[1] Consequently, the development of efficient and selective fluorination and perfluoroalkylation methods constitutes an important yet highly challenging field in modern organic synthesis.[2] While radical trifluoromethylation[3] has emerged as a powerful strategy to introduce the CF_3 group, analogous difluoromethylation[4] and monofluoromethylation[5] have received much less attention. This is quite surprising given that the CF_2H and CH_2F motifs frequently appear in marketed drugs and are known to serve as bioisosteric replacement of various functional groups to improve therapeutic properties.[6] This disconnect has been somewhat addressed, as shown by us[7] and others[8] in a series of transformations involving styrene-type substrates as radical acceptors. However, related processes employing other kinds of unsaturated systems remain scarce. In particular, fluoroalkylative 1,4-difunctionalization of 1,3-enynes, despite recent advances,[9] has been limited to the installation of CF_3 and CF_2R (R = EWG) groups (Scheme 1a).

Building on our previous work on styrenyl substrates,[10] we asked if 1,3-enynes could be employed as potential coupling partners toward acylfluoroalkylation (Scheme 1b). It is noteworthy that, in addition to direct cross-coupling (A) and Heck-type coupling (B), the current transformation may suffer from other side pathways,[11] such as hydrofluoroalkylation (**C**) and the formation of other regioisomers (D and E). Herein, we report the successful development of a three-component acylfluoroalkylation of 1,3-enynes via *N*-heterocyclic carbene (NHC)/photoredox dual catalysis.[12] Remarkably, this protocol is universally applicable to an array of fluoroalkyl sulfinate reagents without the need to reoptimize the reaction conditions.

To achieve this three-component acylfluoroalkylation reaction, we initiated the investigation using 4-phenyl-1,3-enyne **1**, benzoyl fluoride **2a**, and $NaSO_2CF_2H$ **3a** as the model substrates. (Table 1). To our delight, our previous protocol on acylmonofluoromethylation[7b] delivered the desired 1,4-difunctionalization product **4** in 60% yield, along with 15% of regioisomer **5** (Table 1, entry 1). Further screening of other

Scheme 1. Background and Reaction Design



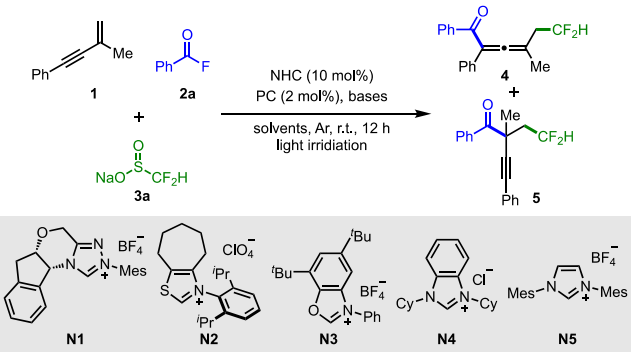
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Table 1. Optimization Studies[a]



entry	photocatalyst	NHC	base	yield (%)[b] 4	yield (%)[b] 5
1	$Ru(bpy)_3(PF_6)_2$	**N1**	Cs_2CO_3	60	15
2	$Ru(phen)_3(PF_6)_2$	**N1**	Cs_2CO_3	52	12
3	$Ir(ppy)_3$	**N1**	Cs_2CO_3	33	18
4	[Ir][c]	**N1**	Cs_2CO_3	75	17
5	3CzEPAIPN	**N1**	Cs_2CO_3	28	7
6	Eosin Y	**N1**	Cs_2CO_3	48	12
7	Rose bengal	**N1**	Cs_2CO_3	35	8
8	[Ir][c]	**N2**−**5**	Cs_2CO_3	<30	<10
9	[Ir][c]	**N1**	CsF	4	1
10	[Ir][c]	**N1**	K_2CO_3	76	17
11	[Ir][c]	**N1**	Na_2CO_3	58	12

[a]General conditions: **1** (0.1 mmol), **2** (0.25 mmol), **3** (0.15 mmol), NHC catalyst (10 mol %), photocatalyst (2 mol %), base (0.2 mmol), acetone (0.1 M), Ar, r.t., 12 h, 50 W blue LEDs. [b]GC yield using diethyl phthalate as the internal standard. [c][Ir] = $[Ir(dF(CF_3)ppy)_2(dtbbpy)](PF_6)$.

Ru- and Ir-based photocatalysts revealed $[Ir(dF(CF_3)ppy)_2(dtbbpy)](PF_6)$ to be the most efficient one (entries 2−4), while organic photocatalysts were found to be inferior (entries 5−7). Different classes of NHC catalysts, including thiazolium (**N2**), oxazolium (**N3**), and imidazolylidenes (**N4** and **N5**), were also examined, although no improvement in yield was observed (entry 8). Further evaluation of the reaction parameters revealed K_2CO_3 as the optimal base (entry 9−11) and acetone as the superior solvent (see Table S4), while changing the concentration, light source, and intensity did not lead to higher yields (see Tables S5−7). Control experiments demonstrated that the reaction did not occur in the absence of catalyst, base, or light irradiation (Table S8).

With the optimized conditions in hand, we surveyed the generality of this three-component coupling (Scheme 2). Thus, electronically diverse 4-aryl-1,3-enynes bearing various functional groups were found to be viable substrates to afford the desired products **6**−**20** in moderate to good yields, including useful synthetic handles such as halides (**10** and **11**) and cyanide (**14**). In the case of a nitro substituent, we observed exclusive 1,4-hydrodifluoromethylation (**15**), while a formyl group shut down the reaction (**18**). Furthermore, heteroarenes underwent a smooth reaction in good yields (**21**−**22**). Notably, 4-alkyl-1,3-enynes could also be employed in this transformation (**23**−**24**), albeit in diminished yields. Besides, reactions using monosubstituted (**25**) and internal 1,3-enynes (**26**) proceeded efficiently, thus offering variability in the substitution pattern of the product allenes. Next, the scope of acyl fluorides was evaluated. Electronic and steric perturbations were generally tolerated (**27**−**38**). In particular, bromo- (**29**) and even iodo-substituted (**35**) acyl fluorides furnished the respective products in good yields. Though less efficient, a benzofuran motif could also be incorporated (**39**). Excitingly, we found that, in addition to acyl fluorides, dicarbonates and acid anhydrides could also participate in the reaction toward products containing an ester and an aliphatic ketone moiety, respectively (**40** and **41**), thus representing a substantial expansion of chemical space beyond (hetero)aryl substituents.

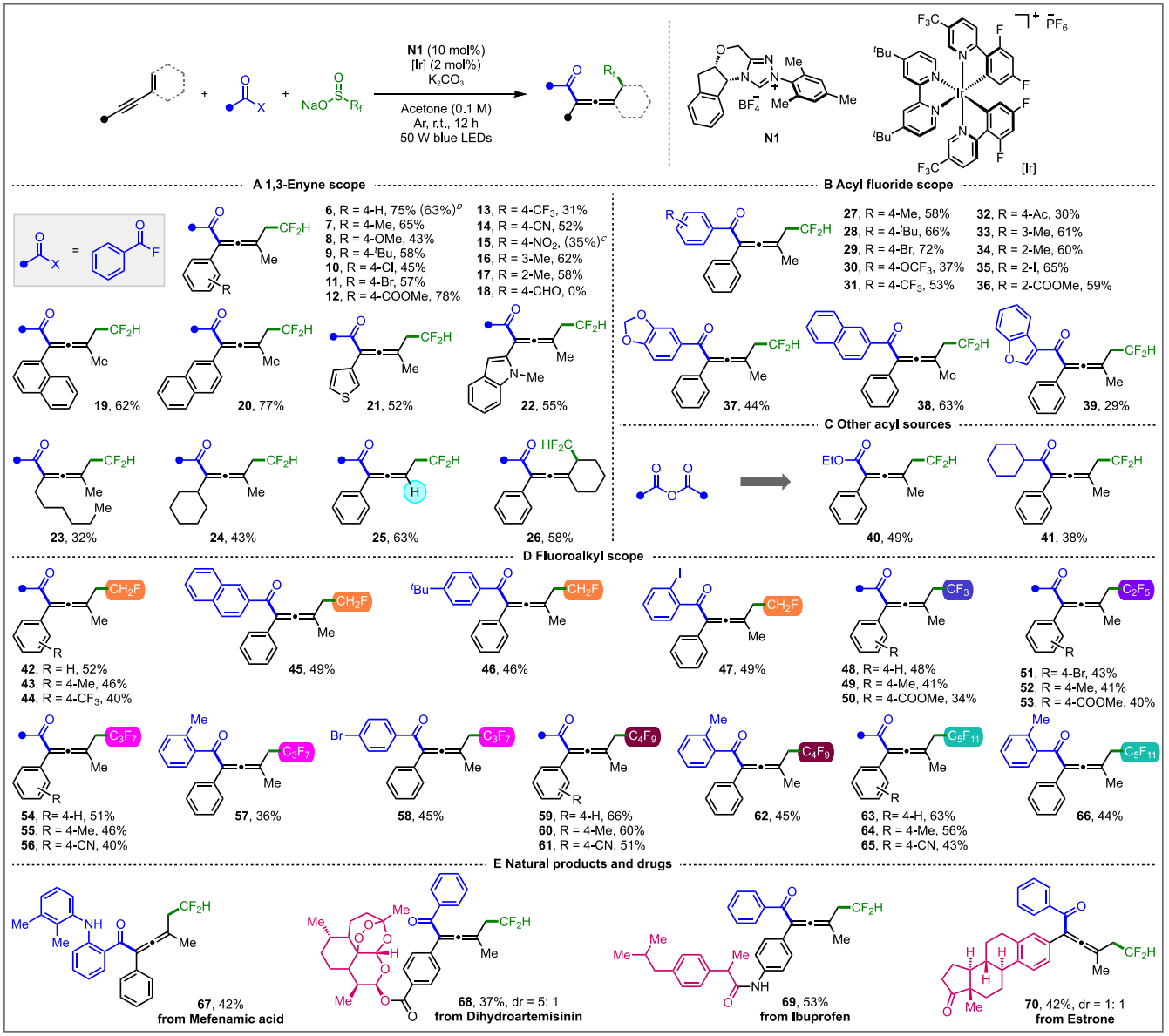
To further explore the generality of this protocol, we synthesized several fluoroalkylsulfinate reagents ($NaSO_2R_f$) and subjected them to our reaction. To our delight, the incorporation of CH_2F (**42**−**47**), CF_3 (**48**−**50**), C_2F_5 (**51**−**53**), C_3F_7 (**54**−**58**), C_4F_9 (**59**−**62**), and C_5F_{11} (**63**−**66**) groups via the current 1,4-difunctionalization cascade could be achieved toward a wide array of fluoroalkylated products. Similar to previous cases, no significant influence on the reaction outcome was observed upon the modification of the arene electronics. Thus, this method offers straightforward access to various fluoroalkylated allenic compounds without the need to adjust the reaction conditions to accommodate different fluoroalkylating reagents.

Our protocol can be applied to the late-stage functionalization of pharmaceutical or bioactive compounds. For instance, acyl fluoride derived from mefenamic acid (**67**) as well as 1,3-enyne derivatives of dihydroartemisinin (**68**), ibuprofen (**69**), and estrone (**70**) were found to be efficient coupling partners.

Several experiments were conducted to gain some insight into the mechanism (Scheme 3). In the radical trap experiments, the addition of 2,2,6,6-tetramethylpiperidinyloxy (TEMPO) or diphenyl ethylene (DPE) to the standard conditions effectively inhibited the acyldifluoromethylation process. The formation of adducts **71**−**73** indicated the presence of fluoroalkyl radicals in the process (Scheme 3a). Further evidence was obtained from EPR studies, in which the

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Scheme 2. Scope of Acylfluoroalkylation of 1,3-Enynes[a]



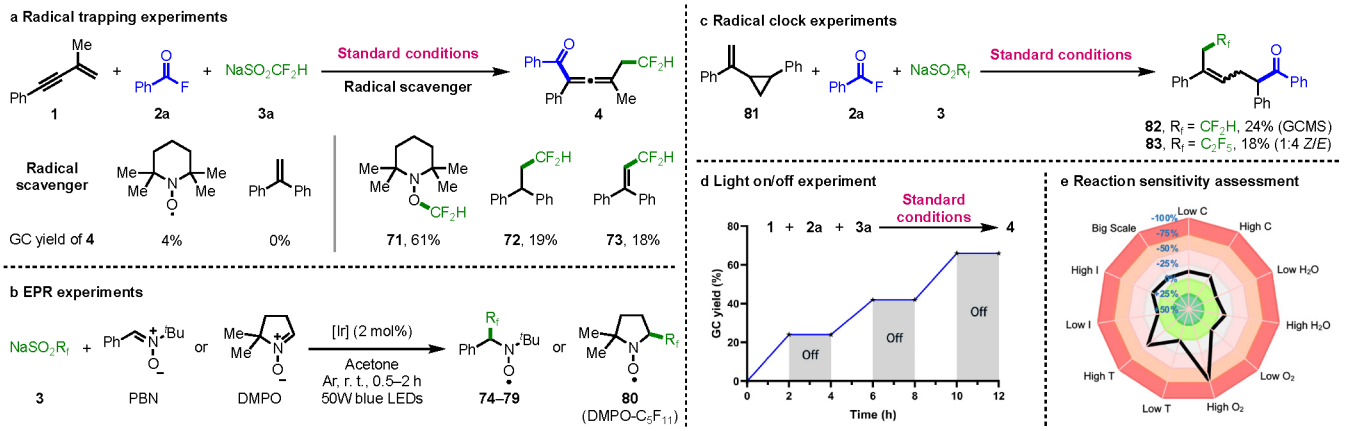
[a]General conditions: 1,3-enyne (0.2 mmol), acyl fluoride (0.5 mmol), $NaSO_2R_f$ (0.3 mmol), **N1** (10 mol %), $Ir[dF(CF_3)ppy]_2(dtbbpy)PF_6$ (2 mol %), and K_2CO_3 (0.4 mmol) in 2 mL of acetone at room temperature under 50 W blue LED irradiation under argon for 12 h. Isolated yield. [b]2 mmol scale. [c]1,4-Hydrodifluoromethylated product was isolated exclusively.

spin traps successfully captured all fluoroalkyl radicals (**74**–**80**) generated from the corresponding sulfinate reagent (Scheme 3b and Supporting Information), as well as a radical clock experiment of **81** leading to ring-opened products **82** and **83** (Scheme 3c). In order to identify the possible intermediates of the reaction, benzoic anhydride **2b**, benzoyl imidazolium **2c** and benzoyl thiazolium **2d** were subjected to the standard conditions (see the Supporting Information). It was found that compounds **2b** and **2c** could lead to product **4**, suggesting their plausible intermediacy. In contrast, **2d** failed to deliver the product, which might explain the low efficiency of catalyst **N2** (cf. Table 1). Light on/off experiments of the model reaction showed that constant irradiation was required, indicating that a radical chain mechanism is less likely (Scheme 3d). Stern–Volmer luminescence quenching studies revealed **3a** to be the primary quencher of photoexcited $Ir[dF(CF_3)ppy]_2(dtbbpy)PF_6$, while 1,3-enyne **1a** and benzoyl fluoride **2a** displayed a negligible effect (see the Supporting Information). Finally, a reaction condition-based sensitivity assessment showed that concentration (C), water content, and light intensity (I) had little effect on the reaction outcome. However, higher temperature (T) and oxygen levels led to significant drops in yield. Therefore, temperature control and careful degassing of the reaction mixture are crucial to the reproducibility of the reaction (Scheme 3e).

Based on our mechanistic studies and literature reports, we proposed the following mechanism (Scheme 4). First, photoexcited Ir(III)* is reductively quenched by $NaSO_2R_f$ **3**, generating the corresponding R_f radical **A** and Ir(II). The former undergoes radical addition to 1,3-enyne **1** to provide a tertiary propargyl radical intermediate **B**, which can isomerize to relatively stable allenyl radical **C**.[9i,13] As an off-cycle process,

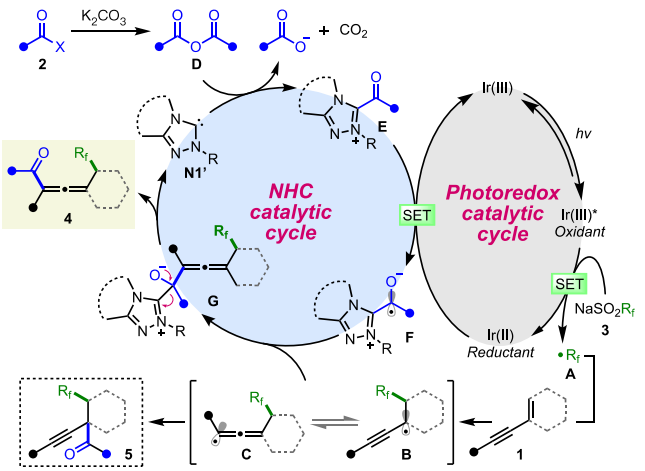
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Scheme 3. Mechanistic Studies



acyl donor **2** reacts with K_2CO_3 to generate anhydride **D**, which then enters the NHC catalytic cycle upon reaction with **N1′** (cf. Supporting Information). The resultant azolium **E** is reduced by Ir(II) to generate ketyl radical **F** while closing the photoredox cycle. Intermolecular radical−radical cross-coupling of persistent radical **F** and allenyl radical **C** leads to NHC-bound intermediate **G**. Finally, elimination leads to the fluoroalkylated allenyl ketone product **4** and regenerates **N1′**. Alternatively, propargyl radical **B** can undergo coupling with **F** to form 1,2-addition side product **5**.

Scheme 4. Proposed Mechanism



In summary, we have developed an efficient three-component acylfluoroalkylation of 1,3-enynes via NHC/PC cooperative catalysis. Various 1,3-enynes and acyl fluorides are compatible, and a range of functional groups can be tolerated. Seven types of fluoroalkyl moieties (CH_2F, CF_2H, CF_3, C_2F_5, C_3F_7, C_4F_9, and C_5F_{11}) can be introduced using this approach. The synthetic utility of the current method is illustrated through late-stage modification of diverse bioactive molecules.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c01372.

General procedure for the cooperative catalytic reaction; optimization of reaction conditions; mechanistic studies; characterization of products; and ^1H, ^{13}C, and ^{19}F NMR spectra of products (PDF)

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Notes

The authors declare no competing financial interest.

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